



Effect of tris(methoxy diethylene glycol) borate on ionic conductivity and electrochemical stability of ethylene carbonate-based electrolyte

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ABSTRACT

The effect of tris(methoxy diethylene glycol) borate (TMDGB) on the coordination structure between ethylene carbonate (EC) solvents with high permittivity and ClO_4^- anions has been investigated by using a Fourier transform infrared (FT-IR) spectroscopy. The results of FT-IR analyses manifested that the boron atom of TMDGB anion receptor forms the complex with ClO_4^- anions. Even though Lewis acid–base interaction between the TMDGB anion receptor and ClO_4^- anions in the electrolyte solution lead to the prominent enhancement of both the dissociation degree of lithium salts and the lithium ion transference number, the ionic conductivity of the EC-based electrolyte solution decreased due to the trap of ClO_4^- anions by introducing the TMDGB anion receptor.

The electrochemical stability of gel polymer electrolyte based on semi-interpenetrating network (IPN) structure with tris(pentafluoro phenyl) borane (TPFPB) or TMDGB anion receptor was obviously improved.

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1. Introduction

Various advanced electrolyte systems have been widely studied to enhance the cyclic performance and safety of rechargeable lithium batteries [1–5]. One of the interesting materials in various advanced electrolyte systems is an anion receptor, which increases both the dissociation degree of lithium salts and the lithium ion transference number of electrolyte solutions [6]. An anion receptor is additionally introduced into the electrolyte solution, which typically consists of lithium salts dissolved in an organic solvent or mixture of organic solvents. To accomplish a complex with salt anions, most of the reported host molecules contain Lewis acid metal centers such as an electron deficient B atom coordinating with electron withdrawing groups [7,8]. The electrolyte solution with lithium hexafluorophosphate (LiPF_6), which the current lithium-ion industry uses exclusively, shows thermal instability. Recently, Herstedt and McBreen et al. reported that tris(pentafluorophenyl) borate (TPFPB) as an anion receptor improved the thermal stability of lithium ion cells containing LiPF_6 -based electrolyte solution [9–11]. It is believed that this behavior is obviously linked to Lewis acid–base interaction between boron atoms of TPFPB anion receptor and PF_6^- anions. Boric acid esters

of glycol solvents, which stabilize alkali metals against corrosion and enhance the anion salvation, were reported [12]. Wakiyara et al. have suggested that the poly(ethylene glycol) borate interacts with ClO_4^- in $\text{Mg}(\text{ClO}_4)_2$ -containing polymer electrolyte [4]. It is well known that the vibrational spectrum of a molecule is markedly affected by environmental changes and in favorable cases the band splitting is used to quantify the new conformational arrangement. The wide use of vibrational spectroscopy is justified by the amount of structural information, which can be readily obtained.

In this work, our group attempted to elucidate the effect of tris(methoxy diethylene glycol) borate (TMDGB) anion receptor on the coordination structure between EC molecules and ClO_4^- anions by using a Fourier transform infrared (FT-IR) spectroscopy. Ionic conductivities of EC-based electrolyte solutions and the anodic limit of gel polymer electrolytes having the TMDGB or TPFPB anion receptor were investigated.

2. Experimental

2.1. Chemicals

Tris(pentafluorophenyl) borane (Aldrich), di(ethylene glycol) methyl ether (Aldrich), $\text{BH}_3 \cdot \text{THF}$ complex (Aldrich), 1,6-hexanediol diacrylate (Aldrich), and methyl benzoylformate (Aldrich) were used as received. Ethylene carbonate (EC, Merck) and propylene carbonate (PC, Merck) were dried over 4 Å molecular sieves prior

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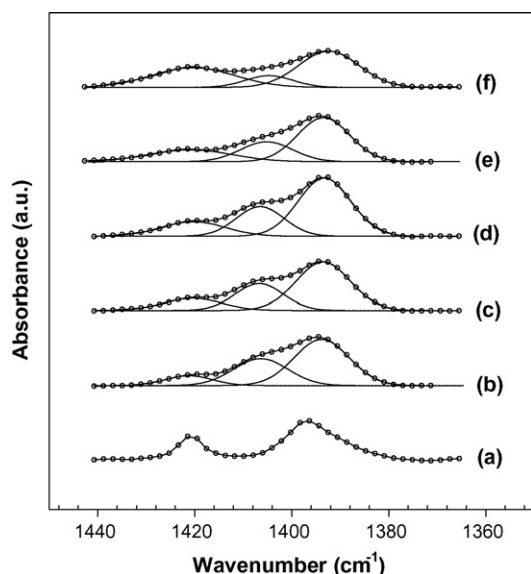


Fig. 1. FT-IR spectra for the CH_2 wagging mode of EC in (a) pure EC, EC/1 M LiClO_4 (b) without TMDGB, with (c) 0.05 mol TMDGB; (d) 0.1 mol TMDGB, (e) 0.2 mol TMDGB, and (f) 0.7 mol TMDGB.

to use. LiClO_4 (Aldrich) was dried in a vacuum oven at 120°C before being added to organic solvents. Tetrahydrofuran (THF), which was used as a solvent for the synthesis of TMDGB and the preparation of gel polymer electrolytes, was dried with the sodium–benzophenone complex.

2.2. FT-IR measurements

FT-IR spectra of the electrolyte solutions were recorded from transmission measurements under a nitrogen atmosphere in a Bruker Tensor 27 spectrometer with a spectral resolution of 4 cm^{-1} in the vibrational frequency range of $400\text{--}4000\text{ cm}^{-1}$.

2.3. DSC measurements

Thermal analyses of electrolyte solution and gel polymer electrolyte were conducted by using a differential scanning calorimeter (DuPont TA 2000 DSC). Each sample was scanned at a heating rate of $10^\circ\text{C}/\text{min}$ within an appropriate temperature range under nitrogen atmosphere.

2.4. Preparation of gel polymer electrolyte

A copolymer of vinylidene fluoride and hexafluoropropylene (Kynar 2801, Atochem Co.) and curable crosslinking agent (1,6-hexanediol diacrylate) were completely dissolved in THF, and then 3 wt% methyl benzoylformate as a photoinitiator and 150 wt% EC/PC/1 M LiClO_4 based a polymer matrix was added in this homogeneous solution. Blend ratio of Kynar 2801 and 1,6-hexanediol diacrylate was 50/50 (w/w). A curable mixed solution was cast on the glass plate and after solvent drying, was irradiated with UV light for 3 min in Ar atmosphere. The thickness of the resulting gel polymer electrolyte based on semi-interpenetrating network (IPN) structure was approximately $100\text{ }\mu\text{m}$. This gel polymer electrolyte based on semi-IPN structure has both suitable flexibility and good mechanical strength.

2.5. Electrical measurements

The ionic conductivities of gel polymer electrolytes were obtained from bulk resistance measured by a.c. complex impedance analysis using a Solartron 1287 frequency response analyzer (FRA) over the frequency range of 10 Hz to 1 MHz.

Lithium transference number was determined by DC polarization/AC impedance combination method [13]. A constant polarization of 10 mV was applied to the cell for 1 h. The transference number of lithium cation was calculated by the relation, $t_{\text{Li}^+} = [I_s(\Delta V - I_0 R_0)] / [I_0(\Delta V - I_s R_s)]$, where I_0 and I_s are the currents at the initial and steady-state, R_0 and R_s are the interfacial resistances at initial and steady-state, respectively.

The electrochemical stability of gel polymer electrolytes was determined by linear sweep voltammetry, using stainless steel as a working electrode and lithium as a reference electrode.

3. Results and discussion

3.1. Effect of TMDGB anion receptor on coordination state between EC molecules and ClO_4^- anions

Tris(methoxy diethylene glycol) borate was synthesized by the dehydrocoupling reaction of a hydroxyl group in di(ethylene glycol) methyl ether with a proton in $\text{BH}_3/\text{tetrahydrofuran}$ complex as described in our prior study [14].

Fig. 1 presents the FT-IR spectra of the region corresponding to the CH_2 wagging mode of EC molecules as a function of TMDGB concentration. If interaction occurs in a material, a peak corresponding to a specific functional group in the FT-IR spectrum shifts toward

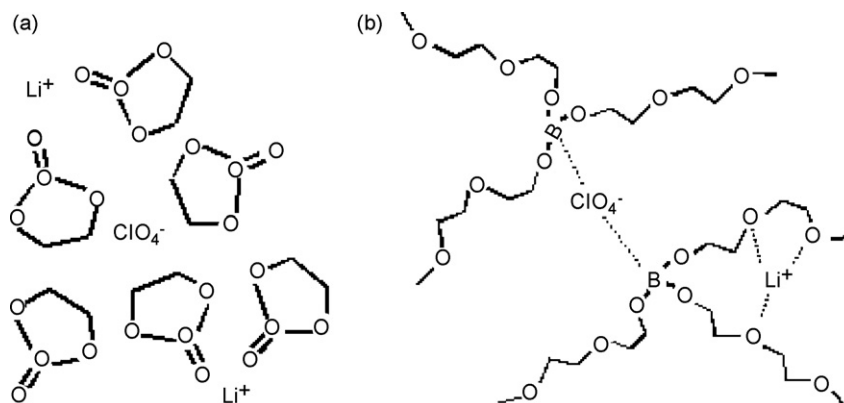


Fig. 2. Schematic illustration for (a) coordination state between EC molecules and ClO_4^- anion and (b) Lewis acid–base interaction between TMDGB anion receptor and ClO_4^- anion.

Table 1

Fraction of solvated EC^a (f_{sec}), apparent solvation number (n), and free ion fraction (f_{ion}) of EC/1M LiClO₄ electrolyte solution

Mole concentration of anion receptor/1 mol LiClO ₄	f_{sec} (%)	n		nf_{ion} (%)
		EC	TMDGB	
0	33	3.2	0	57.0
0.05	28	2.7	0.5	72.6
0.10	26	2.5	0.7	76.0
0.20	24	2.3	0.9	75.2
0.70	12	1.2	2.0	73.0

^a Fraction of EC interacting with ClO₄[−] anions.

either a higher or lower wavenumber or a new peak (or shoulder) appears in the spectrum. The new peak is formed in the region of 1408 cm^{−1} by introducing LiClO₄ into EC solvent, as shown in Fig. 2(a) and (b) [15]. This peak is assigned to the ion–dipole interaction between –CH₂–CH₂– of the EC molecule and ClO₄[−] anions. This ion–dipole interaction is depicted in Fig. 2(a). It is found that the peak intensity of 1408 cm^{−1} region gradually decreases with increasing the TMDGB concentration in EC/1 M LiClO₄ as shown in Fig. 1(c)–(f). Our group could surmise from this result that the TMDGB anion receptor effectively interacts with ClO₄[−] anions and prohibits the interaction of EC molecules with ClO₄[−] anions, as shown in Fig. 2(b). In order to clarify the effect of TMDGB anion receptor on the ion–dipole interaction between EC molecules and ClO₄[−] anions, a curve fitting method was carried out.

Apparent solvation numbers (n) were calculated to analyze quantitatively the interaction between ClO₄[−] anions and EC molecules and these results were summarized in Table 1.

$$n = \frac{C I_S}{C_S(I_S + I_F)},$$

where C is the total molar concentration of EC molecules in the solution, C_S is the salt concentration, I_S is the integrated scattering intensity of solvating molecules, and I_F is the integrated scattering intensity of free EC molecules [16]. The preferential interaction of the electron deficient boron atom of TMDGB anion receptors with ClO₄[−] anions resulted in the reduction of the apparent solvation number of EC molecules in EC/1M LiClO₄ with the increase of the TMDGB concentration as shown in Table 1.

Fig. 3 shows the FT-IR spectra for the region from 610 to 650 cm^{−1} of EC/1 M LiClO₄ with a different TMDGB content. The band around 624 cm^{−1} is assigned to the “free” ClO₄[−] vibrations including free and solvent-separated ions and the one around 632 cm^{−1} corresponds to LiClO₄ ion pair. Dissociation degree of LiClO₄, which was calculated with a rigorous curve fitting method, was summarized in Table 1. The relative fraction of free ions in EC/1 M LiClO₄ electrolyte solution considerably increased from 57% to 72.6% by the addition of 0.05 M TMDGB and then maintained similar value with the increase of TMDGB concentration. It is likely that the enhancement of dissociation degree of EC/1 M LiClO₄ with the TMDGB anion receptor is ascribed to Lewis acid–base interaction between boron atoms of the TMDGB anion receptor and ClO₄[−] anions [4]. In addition to this interaction, it is possible that the dissociation of LiClO₄ is improved by ethylene oxide groups of the TMDGB anion receptor, as shown in Fig. 2(b).

3.2. Ionic conductivities of EC/1 M LiClO₄ and EC/PC/1 M LiClO₄ as a function of TMDGB concentration

Fig. 4 (a) and (b) demonstrate the ion conductivities of EC/1 M LiClO₄ and EC/PC/1 M LiClO₄ electrolyte solution with various TMDGB concentrations. The ionic conductivities of EC/1 M LiClO₄ slightly increased and then decreased with increasing the TMDGB

concentration. The increment in ionic conductivity of EC/1 M LiClO₄ having 0.05 M TMDGB is due to the additional dissociation of LiClO₄ by the strong propensity of TMDGB to form the complex with ClO₄[−] anions, as shown in Fig. 2(b) and Fig. 3. However, higher concentration of the TMDGB anion receptor in the electrolyte solution intends not only to increase the viscosity of the conducting phase but also to restrict the anion migration under an electric field by forming the complex of ClO₄[−] anions with the TMDGB anion receptor. The reduction of mobile anions means the decrease of the concentration of mobile charge carriers under an electric field and results in the decrease of the ionic conductivities of electrolyte solutions. The ionic conductivities of EC/PC/1 M LiClO₄ continuously decreased with increasing the TMDGB concentration. Because the ionic conductivity generally decreases with increasing viscosity, the glass transition temperature (T_g), which means the viscosity of electrolyte solution, was measured by using a differential scanning calorimeter (DSC) and the results were plotted in Fig. 5. Even though T_g of EC/PC/1 M LiClO₄, which reflects the mobility of the free charges, slightly increases from −85 °C (0 M TMDGB) to −83.5 °C (0.7 M), as shown in Fig. 5, the increment is immaterial. From the results of DSC study, it is believed that the reduction of the

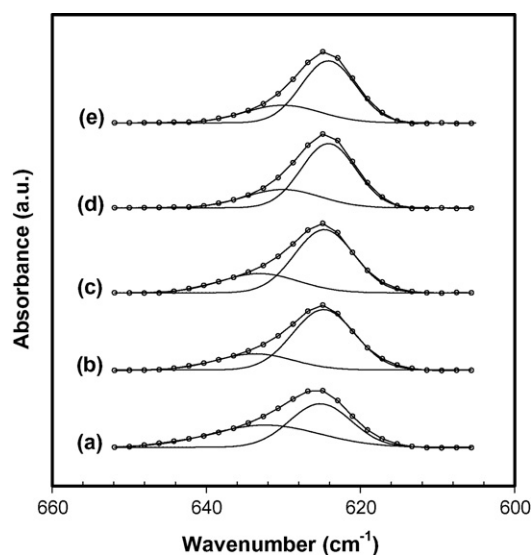


Fig. 3. FT-IR spectra of 610–650 cm^{−1} region of EC/1 M LiClO₄, (a) without TMDGB, with (b) 0.05 mol TMDGB, (c) 0.1 mol TMDGB, (d) 0.2 mol TMDGB, and (e) 0.7 mol TMDGB.

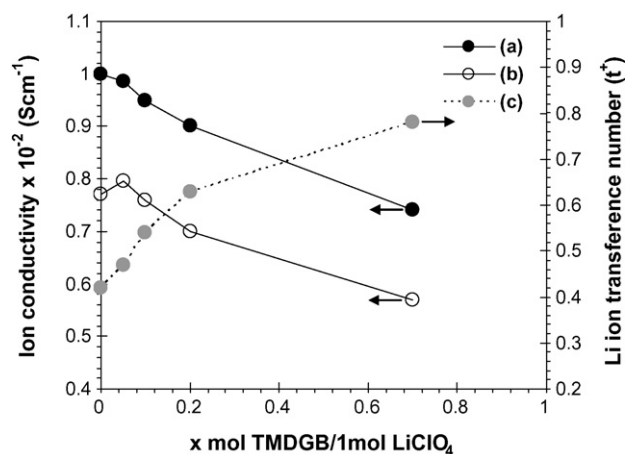


Fig. 4. Ion conductivities for (a) EC/PC (1/1, w/w)/1 M LiClO₄, (b) EC/1 M LiClO₄ at 25 °C, and (c) t_{Li^+} of EC/PC/1 M LiClO₄ with various TMDGB concentrations.

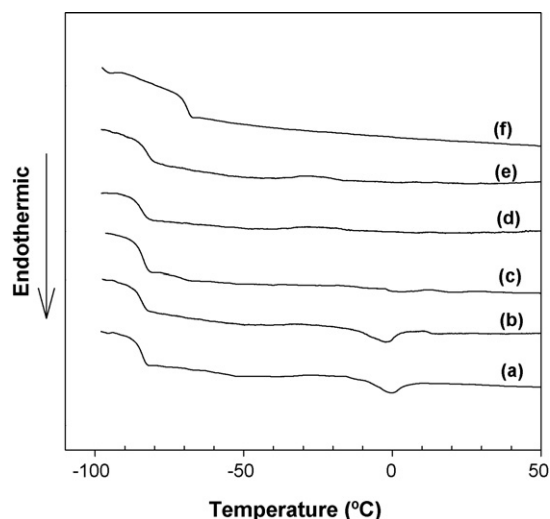


Fig. 5. DSC thermograms of EC/PC/1 M LiClO₄ (a) without TMDGB, with (b) 0.05 mol TMDGB, (c) 0.1 mol TMDGB, (d) 0.2 mol TMDGB, (e) 0.7 mol TMDGB, and (f) pure TMDGB.

ion conductivities of EC/1 M LiClO₄ and EC/PC/1 M LiClO₄ is closely linked to the anion trapping effect by introducing the TMDGB anion receptor.

Since both anions and cations participate in ionic conduction, cationic transport number was considered to elucidate the effect of TMDGB anion receptors on the cationic conductivities. The enhancement of lithium ion transference number of EC/PC/LiClO₄ plotted in Fig. 4(c) indicates that anions were effectively trapped by introducing the TMDGB. The strong interaction between the boron atom of TMDGB anion receptor and ClO₄[−] anions immobilizes anions under an electric field and thereby high lithium ion transference number could be clearly obtained, as shown in Fig. 4(c). Our group could confirm that the diminishment of the ionic conductivity of electrolyte solution with increasing the TMDGB anion receptor is attributed to not its viscosity but the increase of lithium ion transference number owing to the trap of anions.

In our previous study, when the TMDGB anion receptor was introduced into EC/PC/1 M LiClO₄ electrolyte solution, the new peak appeared at −18 ppm in ¹¹B NMR spectrum [14]. It was clearly shown the new peak is originated from the interaction between the boron atom of TMDGB and ClO₄[−] anions, causing more shielding of the boron atom.

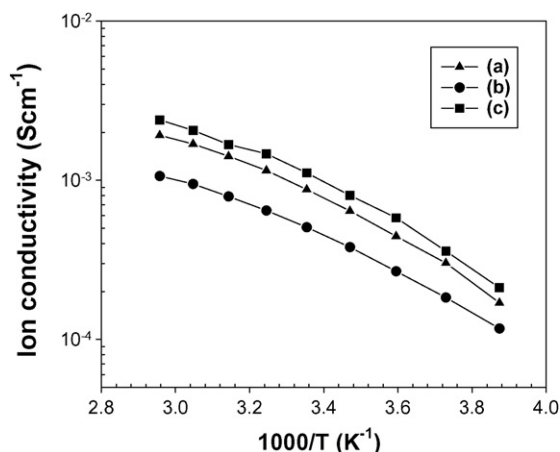


Fig. 6. Arrhenius plots of ion conductivity of gel polymer electrolyte (a) without anion receptor, with (b) 0.02 M TPFPB, and (c) 0.02 M TMDGB.

Table 2

Transference number, T_g , and free ion fraction of gel polymer electrolyte

Anion receptor	Mole concentration of anion receptor/1 mol LiClO ₄	t_{Li^+}	T_g (°C)	Free ion fraction (%)
Without	–	0.57	−81	66
TPFPB ^a	0.2	0.76	−74	73
TMDGB ^b	0.2	0.81	−80	81

^a Tris(pentafluoro phenyl) borane, $M_w = 511$.

^b Tris(methoxy diethylene glycol) borate, $M_w = 497$.

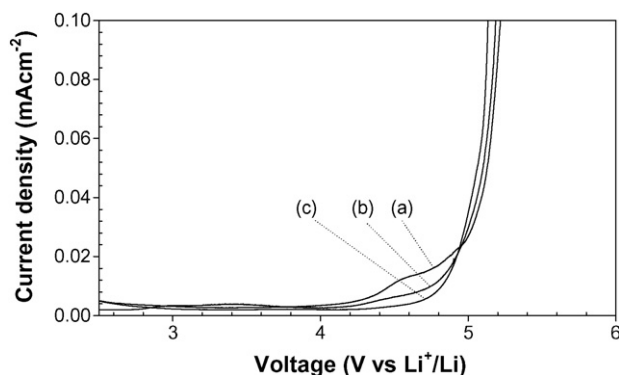


Fig. 7. Current-voltage response of gel polymer electrolyte (a) without anion receptor, (b) with 0.02 M TPFPB, and (c) with 0.02 M TMDGB at 25 °C on the stainless steel electrode as a working electrode (scan rate: 2 mV s^{−1}).

3.3. Ion conductivities and electrochemical stability windows of gel polymer electrolytes with the TMDGB or TPFPB anion receptor

Fig. 6 shows the Arrhenius plots of the ion conductivities for the gel polymer electrolyte based on semi-IPN structure with the TMDGB or TPFPB anion receptor. The gel polymer electrolyte with the TPFPB anion receptor showed much lower ionic conductivities than that of the gel polymer electrolyte without an anion receptor or with the TMDGB anion receptor. This is due to relatively high T_g of −74 °C, causing low mobility of charge carriers, and higher lithium ion transport number, meaning the reduction of mobile anions as shown in Table 2.

Typical current–voltage curves for gel polymer electrolytes with a different anion receptor are presented in Fig. 7. The anodic limit of gel polymer electrolyte was enhanced by introducing TMDGB or TPFPB anion receptor. This indicates that the initiation for the irreversible oxidative decomposition of ClO₄[−] anions in gel polymer electrolyte delays by the strong complexation between the TMDGB or TPFPB anion receptor and ClO₄[−] anions.

4. Conclusions

The TMDGB anion receptor was successfully synthesized by the dehydrocoupling reaction. Lewis acid–base interaction between the TMDGB anion receptor and ClO₄[−] anions in EC/PC/1 M LiClO₄ was obviously confirmed by using FT-IR spectroscopy. The quantitative analyses of FT-IR peaks provided that the coordination degree between anions and polar EC solvent molecules decreased with increasing the TMDGB concentration in electrolyte solution by forming the complex of TMDGB with anions. This complexation could lead to higher lithium ion transference number and the additional dissociation of lithium salts in electrolyte solutions and gel polymer electrolytes. In addition, the electrochemical stability of the gel polymer electrolyte based on semi-IPN structure could be enhanced by the introduction of TMDGB or TPFPB anion receptor.

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